

Product Carbon Footprint Study Report

Product Name: integrated circuit production

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: integrated circuit production

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg integrated circuit production, memory type

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg integrated circuit production, memory type.

2. System Boundary

integrated circuit production, memory type, in China, in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of integrated circuit production produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **1105.452 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

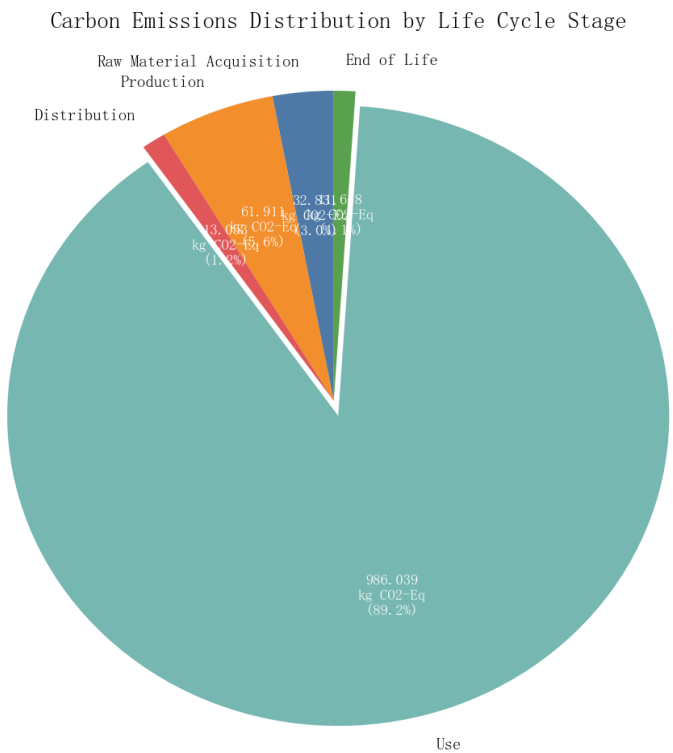


Table 1: Carbon Emissions by Life Cycle Stage

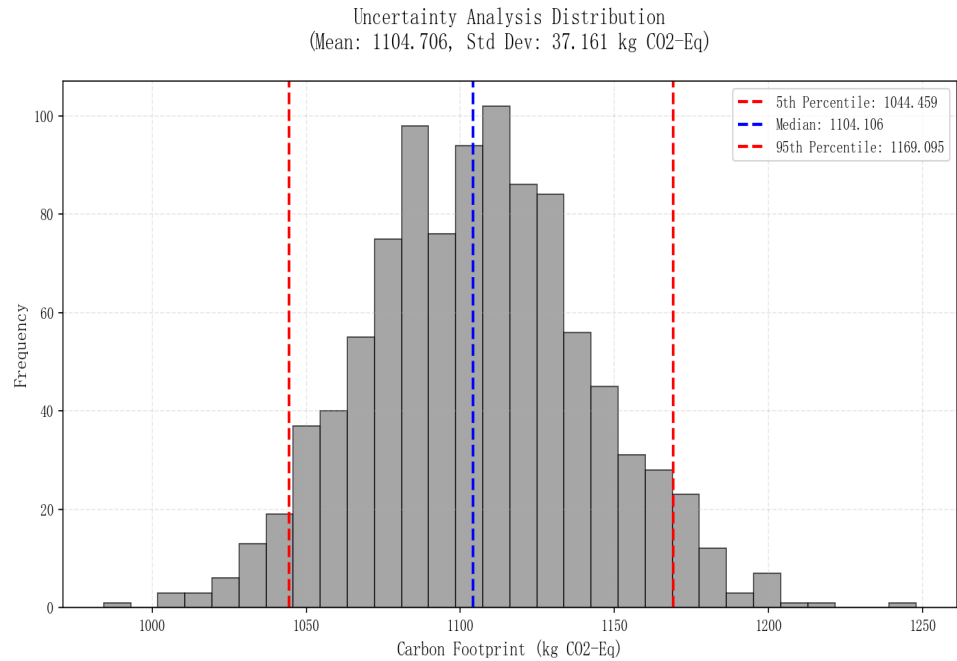
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	32.831	2.97
Production	61.911	5.60
Distribution	13.053	1.18

Use	986.039	89.20
End of Life	11.618	1.05
Total	1105.452	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	1104.706
Standard Deviation	37.161
Coefficient of Variation (CV)	3.36%
Median	1104.106
5th Percentile	1044.459
95th Percentile	1169.095

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the integrated circuit production is calculated to be approximately 1105.452 kg CO₂-Eq per functional unit. This value reflects a significant environmental impact, primarily driven by energy-intensive processes. The life cycle stage analysis reveals that the "use" phase overwhelmingly dominates the carbon footprint, contributing 986.039 kg CO₂-Eq (89.2% of total). This suggests that the product's operational energy consumption over its lifespan is the primary hotspot, far exceeding other stages. In contrast, raw material extraction (3.0%), production (5.6%), distribution (1.2%), and end-of-life (1.1%) have relatively minor contributions, indicating that manufacturing and material sourcing are less critical from a carbon perspective. The major emission sources appear to be electricity or power consumption during product usage, though specific process contributions are unclear due to missing data for top processes (all listed as "Unknown"). This underscores the need for detailed energy profiling of the use phase.

2. Statement of Assumptions and Limitations

This LCA study relies on several key assumptions that may influence results. First, the use phase impact is likely based on an assumed product lifetime and average energy mix, which could vary regionally (e.g., grid carbon intensity). The uncertainty analysis (mean 1104.706 kg CO₂-Eq, CV 3.36%) suggests moderate reliability, but data quality limitations exist—particularly for the top process contributions, which are all marked as "Unknown," indicating gaps in process-level emission factors or allocation. This lack of transparency hampers precise hotspot identification. Additionally, the end-of-life stage (1.1%) reflects assumptions about recycling rates or disposal methods, which may not represent all scenarios. System boundaries appear to cover cradle-to-grave stages, but the "Unknown" processes suggest incomplete inclusion of certain supply chain emissions or ancillary activities (e.g., packaging, transport). The narrow confidence interval (1044.459–1169.095 kg CO₂-Eq) supports robustness, yet missing process data limits actionable insights.

3. Suggestions for Improvement

- ****Improve use-phase efficiency:**** Since the use stage contributes 89.2% of emissions, prioritize designing integrated circuits for lower power consumption (e.g., using advanced node technologies or dynamic voltage scaling). Promote adoption of energy-efficient operation modes in end-user applications.
- ****Optimize production energy mix:**** The production stage (5.6%) offers moderate reduction potential; implement on-site renewable energy (e.g., solar PV) for manufacturing facilities, or purchase green electricity certificates to lower emission intensity.
- ****Enhance data granularity:**** To resolve "Unknown" top process contributions, conduct a detailed life cycle inventory for all input materials and energy flows. Collaborate with suppliers to obtain specific emission factors for raw materials and production steps, replacing generic data.
- ****Extend system boundary:**** Include additional life cycle stages such as product transportation logistics (e.g., air vs. sea freight) and packaging materials, which may be underestimated in current distribution and end-of-life data.

- ****Conduct sensitivity analysis:**** Given the uncertainty CV of 3.36%, perform sensitivity tests on key parameters (e.g., use-phase duration, grid carbon factor) to identify critical factors and prioritize reduction strategies.